

Aryl hydrocarbon receptor literature

Bibliometric Analysis Summary Report

Example configuration for a conservative OpenAlex bibliometric pipeline focused on the aryl hydrocarbon receptor literature.

Query definition

OpenAlex exact-match title/abstract retrieval for "aryl hydrocarbon receptor", "ah receptor", and "dioxin receptor", plus title-level "AHR" hits; local regex validation retained records with explicit AhR naming in the title or abstract-phrase hits supported by biologically relevant title language.

Key corpus statistics

Validated papers	8,291
Time window	1980 to 2026
Abstract coverage	5,290 (63.8%)
Country metadata	7,380 (89.0%)
Disease/application tagged	7,231 (87.2%)
Focus-tagged papers	6,766 (81.6%)
Journals represented	1,609
Countries represented	107

Report scope

The following pages reproduce the generated thesis figures in order. This report is assembled from the current figure files without altering their styling or content.

Corpus Filtering and Retention Flow

Technical overview of retrieval, deduplication, validation, and metadata availability.

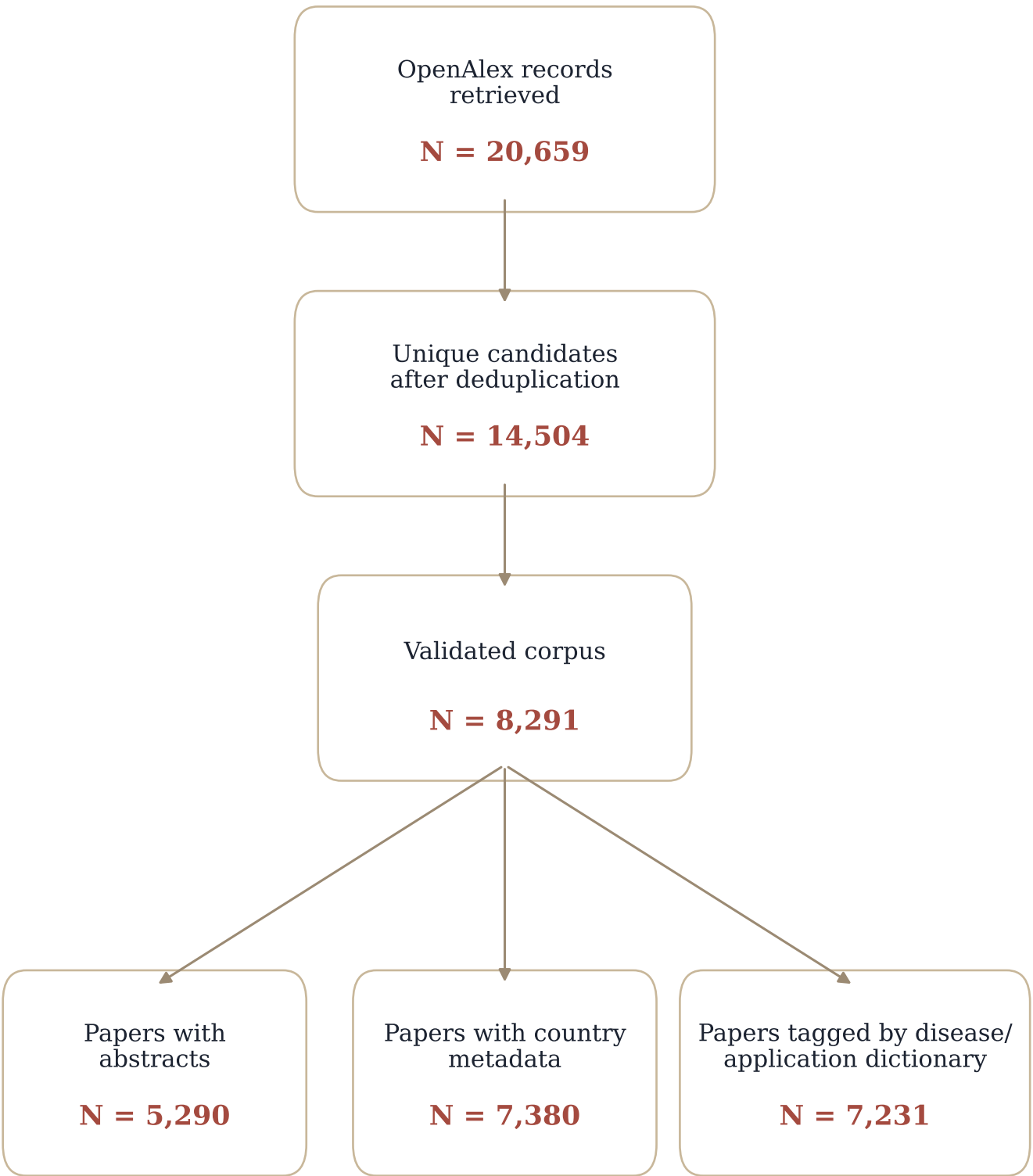
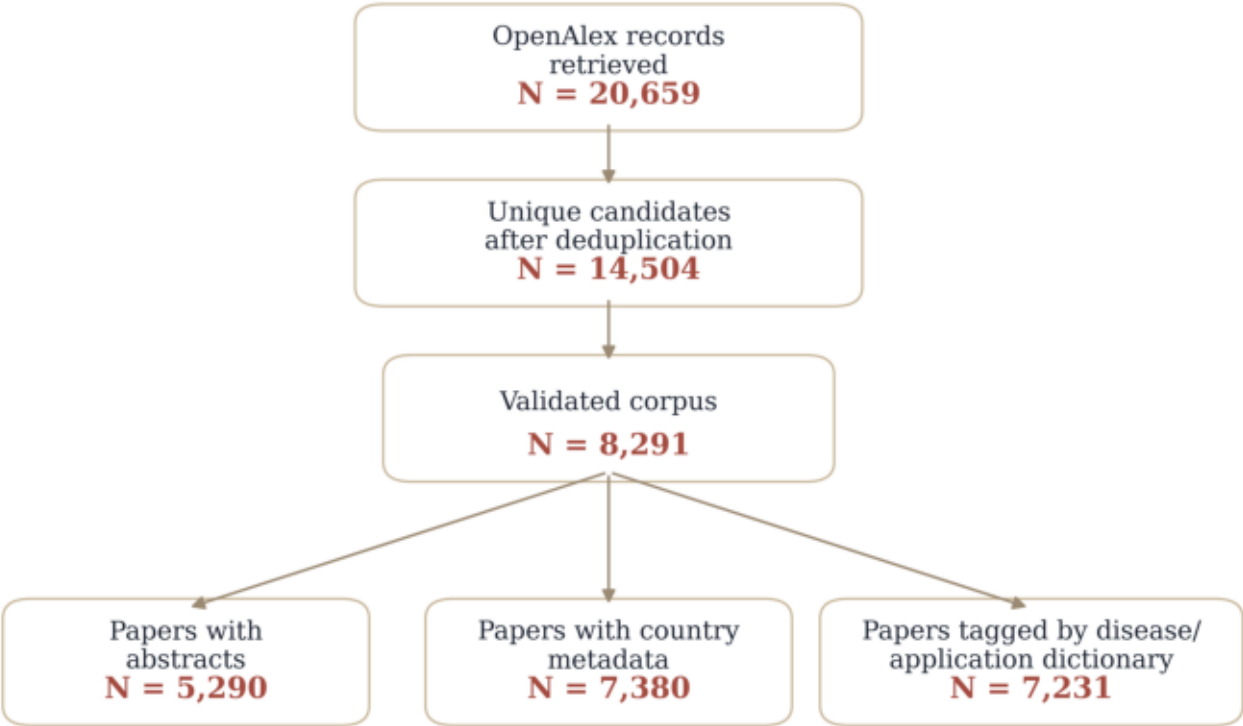


Figure 00. Corpus filtering and metadata retention flow

Corpus Filtering and Retention Flow

Technical overview of retrieval, deduplication, validation, and metadata availability.



Counts are drawn from the current cached retrieval, validated corpus table, and downstream metadata coverage summaries.

Figure 01. AhR literature growth over time

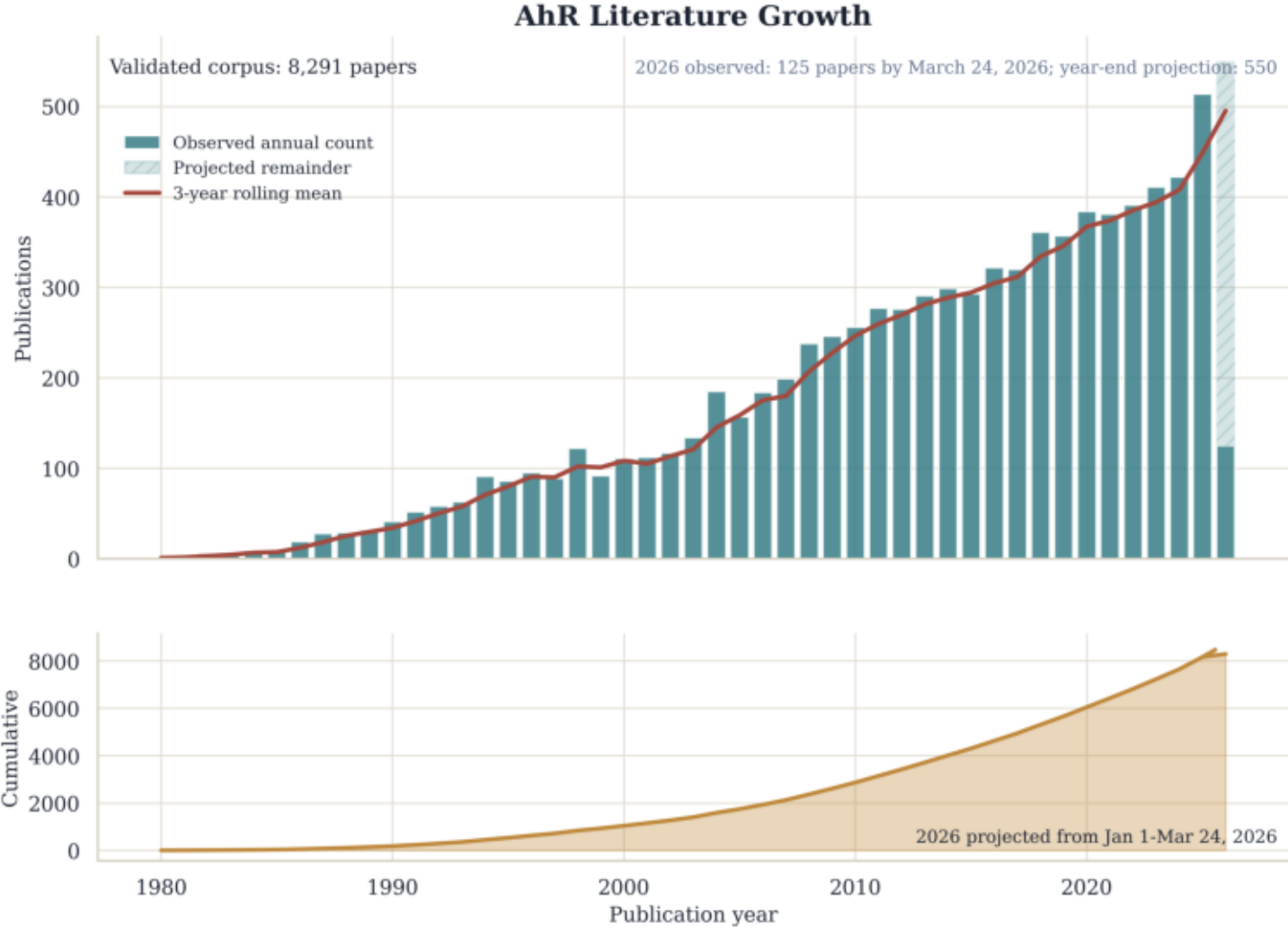


Figure 02. Disease and application distribution across the AhR corpus

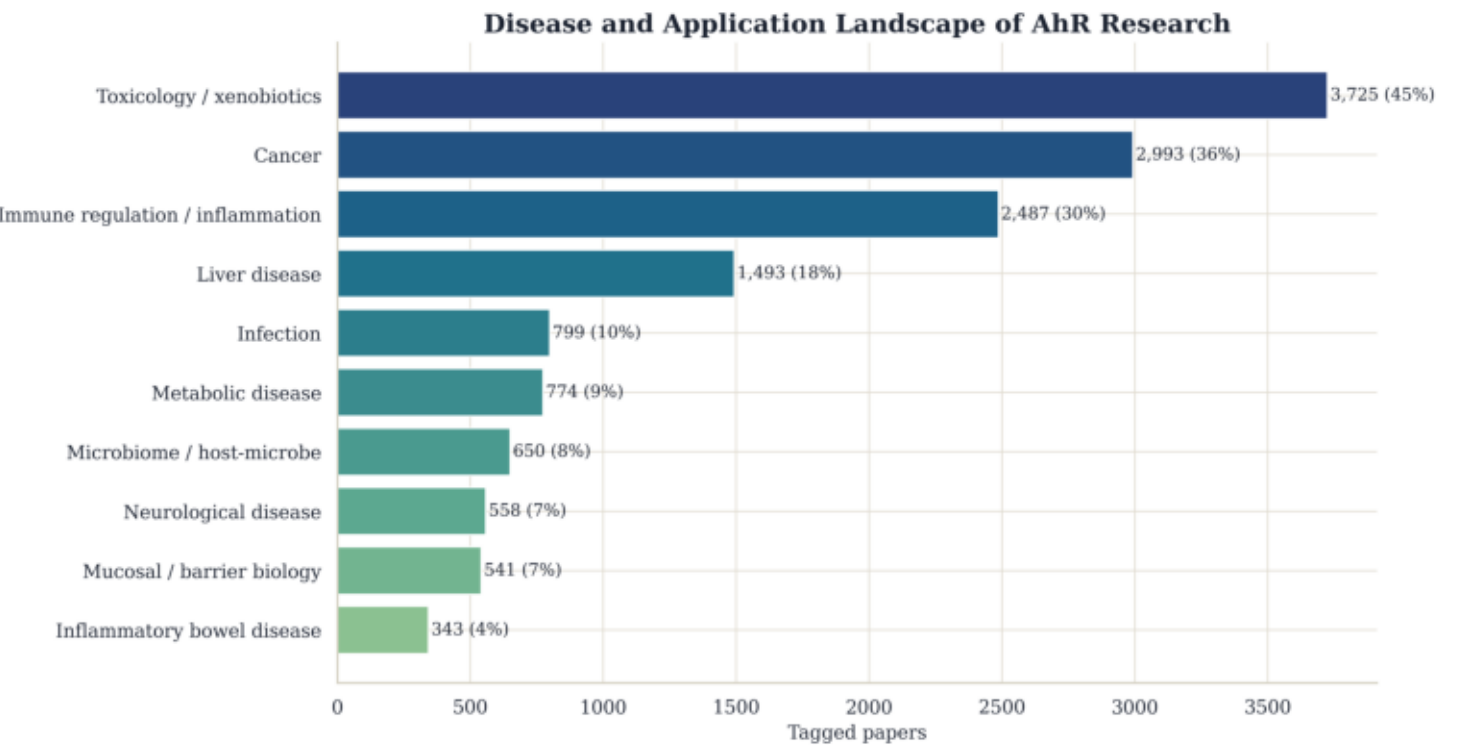
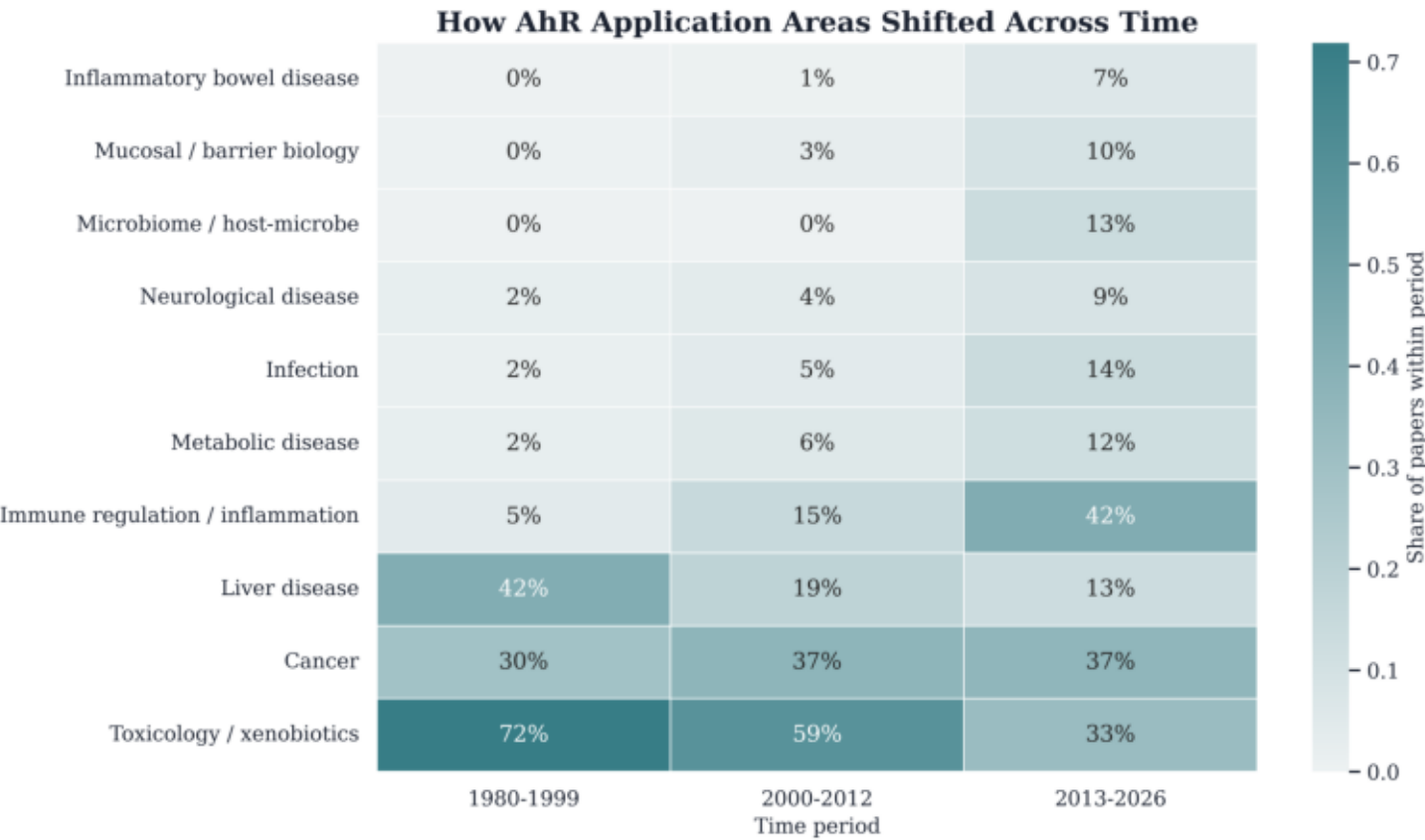


Figure 03. Disease and application trends across AhR field eras



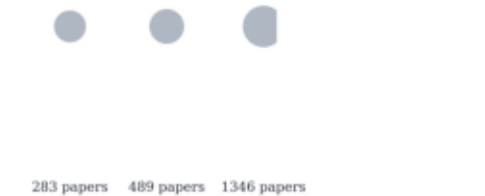
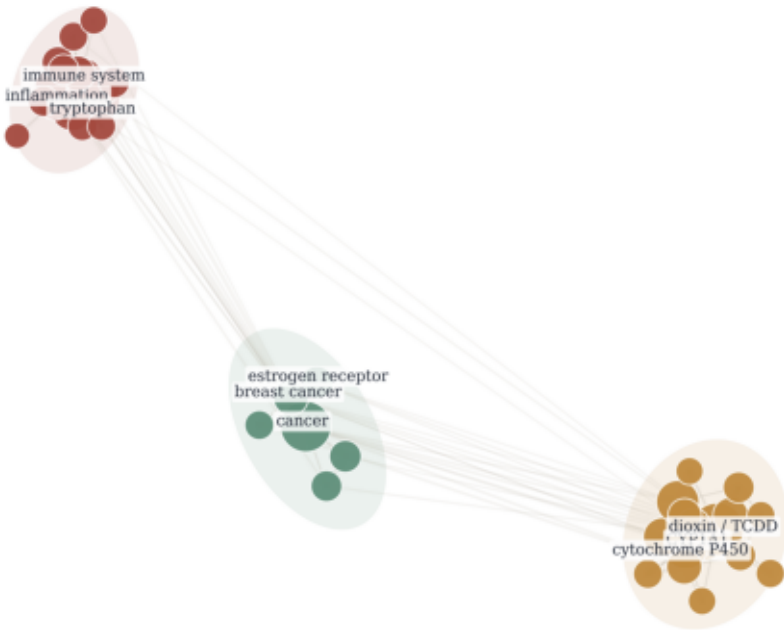
Conceptual Landscape of AhR Research

How to read this map

- Node size: number of papers carrying the concept
- Node color: thematic cluster from Louvain community detection
- Edge width: co-occurrence strength between concept labels

Conceptual Landscape of AhR Research

Curated concept co-occurrence map across the full validated AhR corpus



Thematic clusters

- Cluster 2: Dioxin and xenobiotic toxicology
CYP1A1, dioxin / TCDD, cytochrome P450, CYP1B1, carcinogen
- Cluster 1: Immune-microbiome signaling
immune system, inflammation, tryptophan, gut, indoles
- Cluster 3: Cancer and hormone signaling
cancer, breast cancer, estrogen receptor, carcinogenesis, oxidative stress

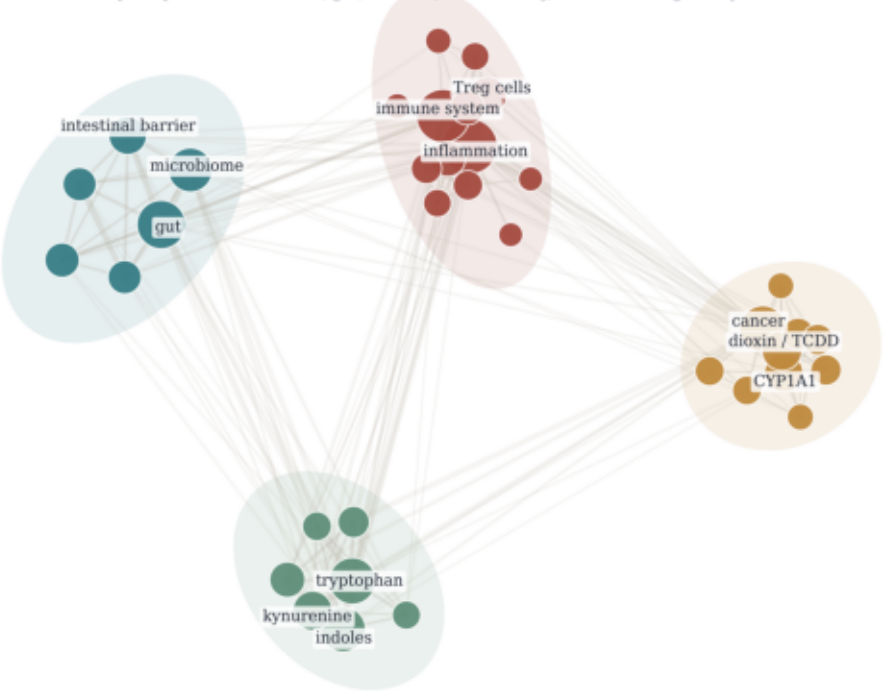
Immune-Microbiome-Barrier AhR Sublandscape

How to read this map

- Node size: number of papers carrying the concept
- Node color: thematic cluster from Louvain community detection
- Edge width: co-occurrence strength between concept labels

Immune-Microbiome-Barrier AhR Sublandscape

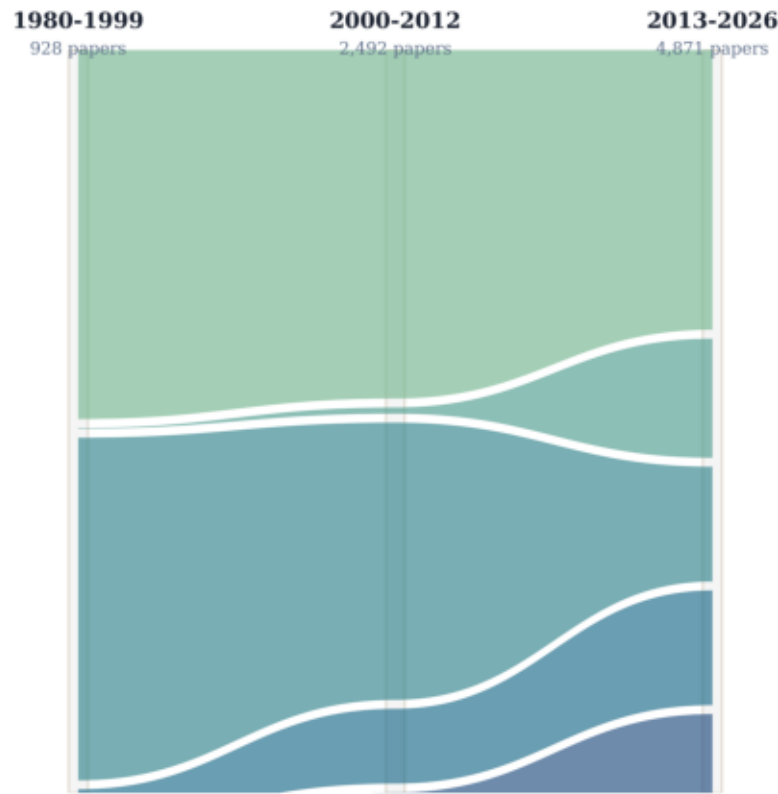
Focused concept map of the microbiome, gut, mucosal, inflammatory, and immunoregulatory AhR literature



Thematic Evolution of AhR Research

Thematic Evolution of AhR Research

Ribbon width shows the share of papers assigned to each displayed theme within each era.



How to read this map

Ribbon color: displayed theme
Ribbon width: within-period theme share
Columns: 1980-1999, 2000-2012, 2013-2026

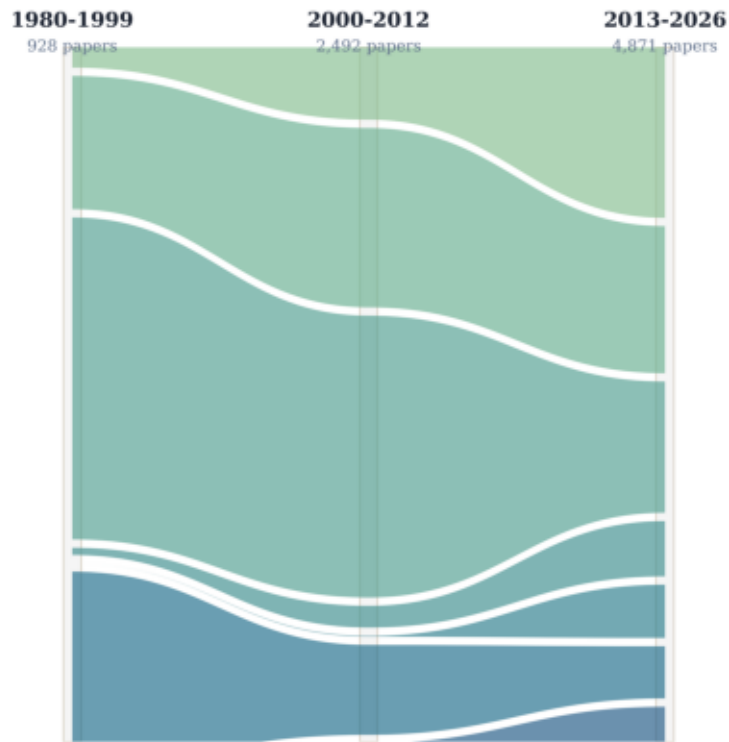
Themes tracked across time

- Environmental toxicology and liver response
Recent-era share: 38%
- Immune-microbiome signaling
Recent-era share: 16%
- CYP1-centered toxicology and transcription
Recent-era share: 15%
- Cancer and hormone signaling
Recent-era share: 15%
- T cell and cytokine regulation
Recent-era share: 15%

Disease and Application Evolution of AhR Research

Disease and Application Evolution of AhR Research

Ribbon width shows the relative composition of the most prevalent disease/application categories within each era.



How to read this map

Ribbon color: disease/application category
Ribbon width: relative share among displayed categories
Same category framework as Figure 03

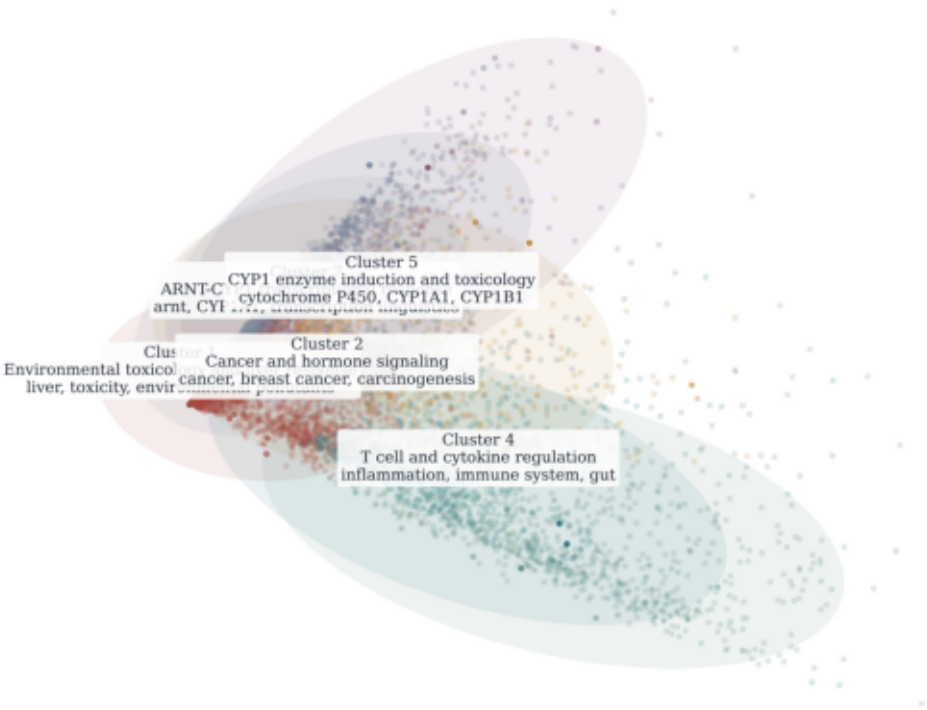
Themes tracked across time

- Immune regulation / inflammation
Recent displayed share: 25%
- Cancer
Recent displayed share: 21%
- Toxicology / xenobiotics
Recent displayed share: 19%
- Infection
Recent displayed share: 8%
- Microbiome / host-microbe
Recent displayed share: 8%
- Liver disease
Recent displayed share: 8%
- Metabolic disease
Recent displayed share: 7%
- Neurological disease
Recent displayed share: 5%

Document Landscape of AhR Research Themes

Document Landscape of AhR Research Themes

Each point is one paper positioned in 2D concept space; colored islands summarize the major thematic clusters.



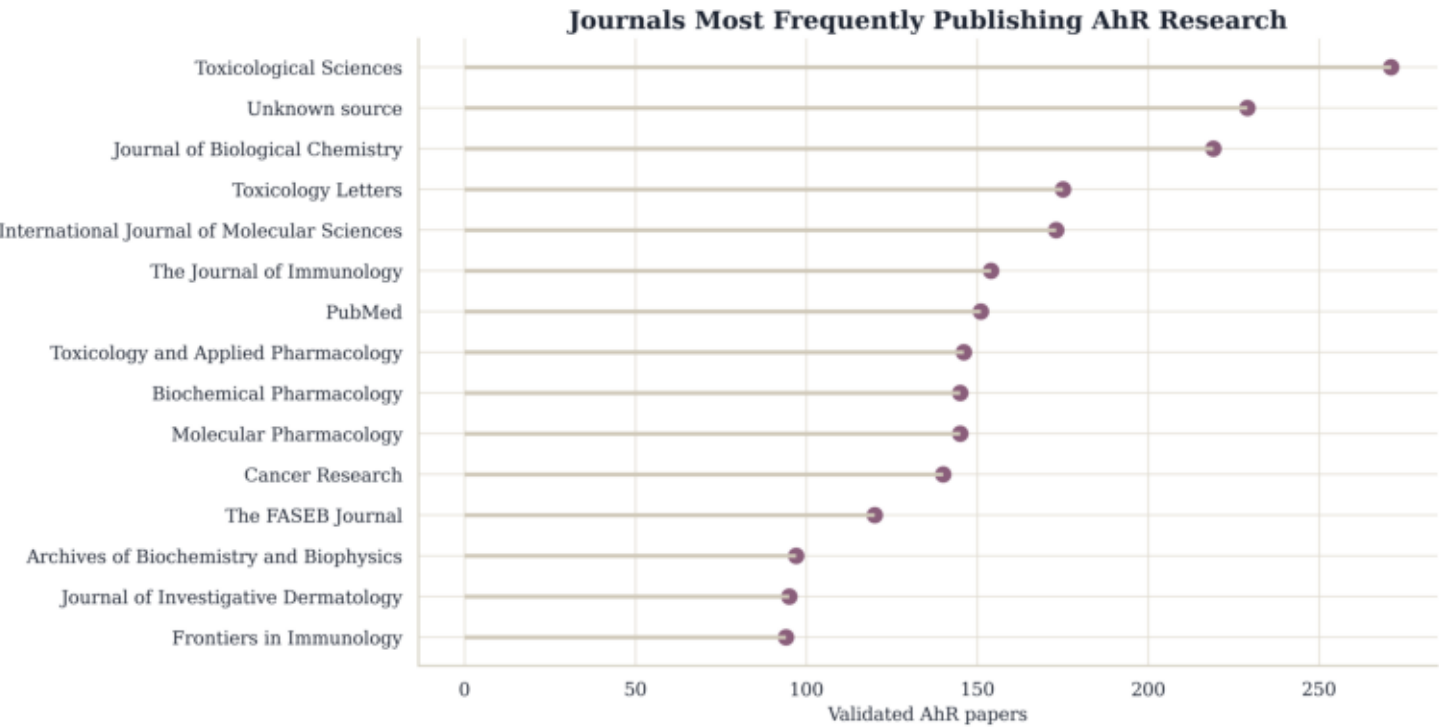
How to read this map

- Each point: one paper
- Color and island envelope: thematic cluster
- Distances: qualitative similarity in concept-profile space

Cluster summaries

- Cluster 1: Environmental toxicology and liver response
3,464 papers | median year 2013
- Cluster 2: Cancer and hormone signaling
1,039 papers | median year 2018
- Cluster 7: ARNT-CYP1 transcriptional response
976 papers | median year 2008
- Cluster 5: CYP1 enzyme induction and toxicology
884 papers | median year 2008
- Cluster 4: T cell and cytokine regulation
874 papers | median year 2020
- Cluster 3: Immune-microbiome signaling
803 papers | median year 2022
- Cluster 6: CYP1 toxicology and carcinogenesis
251 papers | median year 2011

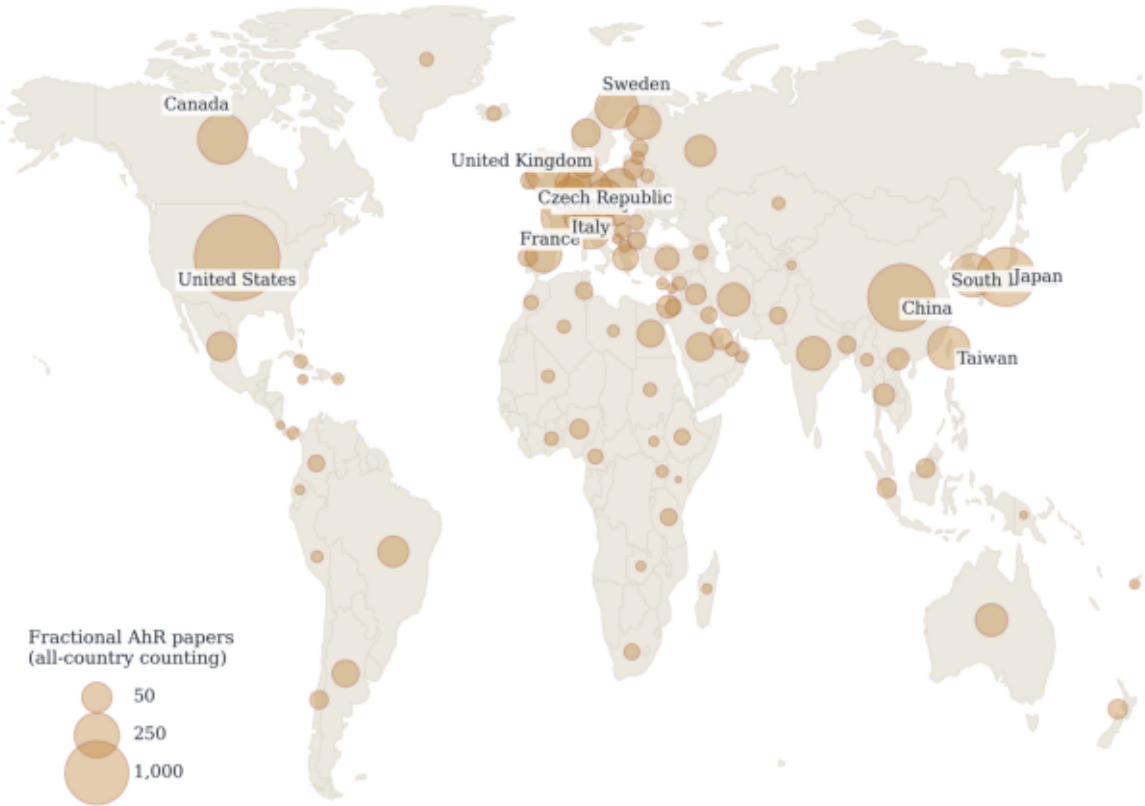
Figure 08. Journals most frequently publishing AhR papers



Global AhR Research Output

Global AhR Research Output

Bubble area shows fractional AhR paper count by country.



Fractional counting distributes each paper across all represented author-affiliation countries. Country metadata available for 7,380 of 8,291 papers (89%).

Global AhR Research Output Per Capita

Global AhR Research Output Per Capita

Bubble area shows fractional AhR papers per million inhabitants.

